Pratik Vimalbalaji

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EDUCATION

Northeastern University — Boston, MA *Expected Dec 2028*

Master of Science in Information Systems

- Relevant Coursework: Data Science Engineering Methods and Tools, Application Engineering and Development

Vellore Institute of Technology — Chennai, India *Jul 2024*

Bachelor of Technology in Computer Science and Engineering (AI & ML)

TECHNICAL SKILLS & CERTIFICATIONS

- Languages & Scripting: **Python**, Java, C++, C, JavaScript, TypeScript, Bash
- ML / AI / Computer Vision: **TensorFlow, LangChain, DSPy**, Keras, scikit-learn, NumPy, OpenCV, NLP
- Web Development: Next.js, React, Shadcn, Tailwind CSS, Stripe, REST APIs, OAuth
- Data, Databases & Cloud: SQL (MySQL, PostgreSQL), **AWS** (S3, EMR), Apache Spark, Hadoop, Apache Hive, Sqoop, Apache Kafka, Databricks, ChromaDB
- Testing & DevOps: Selenium (Hybrid & BDD), Robot Framework, CI/CD, Git, JMeter, Postman, Agile/Scrum
- Certifications: Oracle Cloud Infrastructure Generative AI Professional; Oracle Cloud Infrastructure Data Science Professional; Microsoft Azure AI Fundamentals (AI-900); Microsoft Azure Data Fundamentals (DP-900)

EXPERIENCE

LTIMindtree Pvt. Ltd. — Bengaluru, India *Feb 2025 – Jul 2026*

Software Development Engineer in Test

- Refactored and consolidated the Visual Packaging Systems (VPACS) test suite from **756 feature files to 25**, eliminating redundant scenarios and cutting regression execution time
- Automated generation of Gherkin feature files, step definitions, and manual test cases with GitHub Copilot–powered GenAI testing agents driven by natural-language prompts
- Established Selenium Hybrid and BDD automation frameworks with Robot Framework and CI/CD pipelines for regression testing across Agile sprints, using Postman for API testing
- Led a **10-person team** in LTIMindtree's Full-Stack Apprentice Program, personally ranking in the **top 3 of 60 employees**
- Built a Java-based Library Management System with book checkout, inventory, and member-management features under Agile sprint planning

PROJECTS

AI-Email Bot *Sept 2024 – Feb 2025*

- Built a full-stack email-client clone in Next.js 14 (App Router) with Shadcn/Tailwind CSS, syncing send, receive, and compose via Aurinko, an AI smart-compose copilot, a Cmd-K command bar, and Stripe-powered SaaS subscription billing
- Developed a multimodal RAG pipeline with LangChain that parses uploaded PDF, Excel, Word, and JSON files (text, tables, and images) into embeddings stored in ChromaDB for document-grounded query answering
- Leveraged DSPy to prompt-engineer and optimize a GPT-4o model for the RAG pipeline, improving response accuracy on document-grounded queries

Super Long Range CNN (SLRCNN) for Video Enhancement — *VIT Capstone* *Dec 2023 – May 2024*

- Designed a two-stage CNN pipeline (LRCNN + SRCNN) in TensorFlow/Keras to classify handball actions and enhance frame quality, reaching **91.43% classification accuracy** (training loss 0.3178)
- Implemented video frame extraction, normalization, and sequencing with OpenCV for training-data preparation
- Benchmarked SLRCNN against Mask R-CNN and optical-flow baselines using PSNR/SSIM, raising enhanced-frame PSNR by up to **9.3 dB** over the original low-quality input

Polyp Segmentation Using UNet and ENet — *Best Paper Award, ICRTAC-2023 (IEEE)* *Jul 2023 – Dec 2023*

- Trained UNet and ENet segmentation models in TensorFlow/Keras to localize polyps in endoscopy images, using OpenCV for preprocessing and mask generation
- Ran a comparative study between the two architectures on IoU, accuracy, and inference time, improving segmentation efficiency by **20% over baseline**
- Tuned hyperparameters via Grid Search and Random Search to improve validation accuracy and reduce training time